

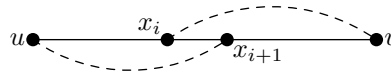
Chapter 8: Hamilton Cycles and Beyond

We define a **Hamilton cycle** as one that contains all the vertices; similarly a **Hamilton path** is one that contains all the vertices.

DIRAC'S THEOREM. *If G is a graph with order $n \geq 3$ and minimum degree $\delta \geq n/2$, then G contains a Hamilton cycle.*

PROOF. Suppose there is a counterexample, meaning a graph with the minimum degree condition that does not have a hamilton cycle. Then consider the counterexample G with the maximum number of edges. Clearly G is not complete; say it is missing edge uv . So consider the graph $G + uv$. By our discussion, this has a Hamilton cycle, and that Hamilton cycle uses edge uv . So in G there is a Hamilton path P from u to v .

Say P is $u = x_1, \dots, v = x_n$. Suppose there is i with $2 \leq i \leq n - 2$ where the pair $Q_i = \{ux_{i+1}, vx_i\}$ of edges both exist; then we get a cycle using all vertices of G .



The union of the Q_i accounts for all the edges incident with u or v , except for ux_2 and vx_{n-1} . Thus the total number of edges incident with them is at most $n - 1$. But that contradicts the minimum degree condition. $\square$

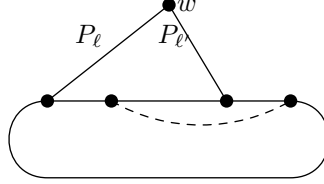
This theorem is best possible. Here are two graphs with minimum degree $(n - 1)/2$ that do not have a Hamilton cycle: $K_{m,m+1}$; or two copies of K_m with one vertex of each identified. There are many extensions. The first was by Ore: if the degree-sum $d(u) + d(v) \geq n$ for all nonadjacent vertices u and v , then the graph has a Hamilton cycle.

CHVÁTAL–ERDŐS. *If G is a graph with order at least 3 and the connectivity is at least the independence number, then the graph has a Hamilton cycle.*

PROOF. If the independence number is 1, then the graph is complete. So assume the independence number $k \geq 2$. In particular this means the connectivity is at least 2. Consider the longest cycle C . Suppose there exists a vertex w outside the cycle. By Menger's Theorem, there are internally disjoint paths $P_1, \dots, P_k$

from w to C . If any two paths meet C at consecutive vertices of C , then one can insert w to get a longer cycle. So assume not.

For each path P_ℓ , let v_ℓ be the vertex of C after the end of P_ℓ going clockwise. We claim that: the set $\{v_\ell\}$ is an independent set. This follows because a chord $v_\ell v_{\ell'}$ would enable a longer cycle using the chord, P_ℓ , and $P_{\ell'}$.



But by the condition of theorem, adding w to $\{v_\ell\}$ yields a set that is not independent. This means that w is adjacent to some v_ℓ , and one can thus insert w using P_ℓ and this edge. This contradicts the claim that C is a longest cycle.

□

For planar graphs, Tutte showed that 4-connected implies the existence of a Hamilton cycle. In general, a necessary condition for a Hamilton cycle is that the removal of any vertex set S leaves at most $|S|$ components. The **toughness** of a graph is defined to be the minimum of the ratio $|S|/k(G - S)$ over all cut-sets S , where $k(G - S)$ is the number of components of $G - S$.

OPEN QUESTION. *Does a sufficiently large toughness guarantee a Hamilton cycle?*

A graph is **pancyclic** if it contains cycles of each length from 3 up to its order. The minimum-degree threshold to guarantee a graph is pancyclic is only slightly larger than Dirac's bound:

MINIMUM DEGREE. *If G is a graph with order $n \geq 3$ and minimum degree $\delta \geq (n + 1)/2$, then G is pancyclic.*

However, if the graph is known to have a Hamilton cycle, then the threshold can be reduced:

BFG. *If G is a graph that contains a Hamilton cycle and a triangle and minimum degree $\delta \geq (n + 2)/3$, then G is pancyclic.*